Lecture 3

How Bacteria Resist Antibiotics

Day 1 · 13:00–13:45 · 45 min

LO4 (part): understand the genetic basis of AMR so that genome results can be interpreted

Four ways to survive an antibiotic

1 · Destroy the drug
Enzymes such as β-lactamases cut penicillins and cephalosporins

2 · Change the target
The drug no longer binds (PBP2a in MRSA; *gyrA* mutations)

3 · Pump it out
Efflux pumps

4 · Keep it out
Fewer or changed porins in Gram-negatives

Where resistance comes from: two routes

ACQUIRED GENE

The bacterium picks up a **new gene** from another bacterium

mecA, *blaCTX-M-15*, *blaNDM-1*

Genome: is the gene present or absent?

CHROMOSOMAL MUTATION

A change in the bacterium's **own gene**

gyrA S83L (ciprofloxacin)

Genome: compare with a reference and look for specific mutations

Plasmids: why resistance spreads so fast

- **Plasmids:** small circles of DNA that move **between bacteria, even between species**
- **Transposons and integrons:** “cut-and-paste” elements that move genes around
- One plasmid can carry **many** resistance genes at once

KEY MESSAGE

A resistant plasmid can spread even when the strain is not spreading. Genomics can tell these apart.

The big four to recognise

mecA
MRSA: resistant to almost all β-lactams

***blaCTX-M* (ESBL)**
Resistant to 3rd-generation cephalosporins such as ceftriaxone

Carbapenemases
KPC, NDM, OXA-48, VIM, IMP, OXA-23: resistant to carbapenems, the “last line”

***gyrA* / *parC* mutations**
Fluoroquinolone (ciprofloxacin) resistance

You don’t need to memorise gene names. The tools report them, and the glossary has a cheat sheet.

Other mechanisms you'll see

Genes	Resistance to
blaSHV, *blaTEM* (ESBL variants)	3rd-generation cephalosporins
aac, *aph*, *ant*, *armA*, *rmtB*	Gentamicin, amikacin
sul1, *sul2*, *dfrA*	Co-trimoxazole
tet(A), *tet(M)*	Tetracyclines

Reading gene names

blaCTX-M-15

bla	CTX-M	15
β-lactamase	family	variant

- aac(6')-Ib-cr: aminoglycoside enzyme; “cr” = also affects ciprofloxacin
- mecA: methicillin resistance

1-MINUTE EXERCISE

What do you predict blaNDM-1 does?

Can the genome predict the AST result?

WORKS VERY WELL

- *S. aureus*
- β-lactams in *E. coli* and *Klebsiella*

WORKS LESS WELL

- Resistance depends on how much a gene is **expressed**
- **Combinations** of weak mechanisms
- **Unknown** mechanisms not yet in databases

KEY MESSAGE

The genome complements AST. It does not replace AST for treating patients.

When the genome and the AST disagree

Genome says	AST says	Possible explanations
Resistance gene	Susceptible	Gene not expressed or broken · lab error · mixed culture
No gene found	Resistant	Unknown mechanism · mutation not in database · low coverage · mixed culture

> Disagreements are **useful**: they point to new mechanisms or lab QC problems. We'll practise this in Lab 4.

AMR tools and databases (preview of Lab 4)

Tool	From	Notes
AMRFinderPlus	NCBI, USA	Genes + mutations; organism-aware
ResFinder / PointFinder	CGE, Denmark	Web tool; predicts phenotype
CARD / RGI	McMaster, Canada	Very comprehensive
ABRicate	Community	Fast screening, several databases

> Tools may disagree slightly. Always report which tool and database version you used.

Summary

- Resistance = **genes** (often mobile) + **mutations**
- Plasmids let resistance spread **between species**
- Genome predictions are powerful but have limits: **always read them next to AST**

> Next: let’s make some genomes (Lab 2).